

SOME VIEWPOINTS ON THE USE OF A COMPUTER FOR CALCULATING PHASE DIAGRAMS

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There are many reasons why simple thermodynamic models are still used extensively for calculating phase diagrams and relating them to thermodynamic properties. More powerful computer software may help to promote a more widespread usage of advanced models. It should be possible to construct software which would allow us to apply new and advanced models without new programming. As a demonstration we describe how a program can be constructed which permits the calculation of phase equilibria under a large number of conditions and which can accept many types of models.

1. Introduction

When Professor Jan Meijering made his now classical contributions to the art and science of phase diagram calculation, he had to make his calculations by hand. It was natural that he chose simple models for the description of the thermodynamic properties of the individual phases. He thus made extensive use of the regular solution model. In spite of this simplification his work was very successful and has inspired a whole generation of scientists to continue using simple solution models. One may wonder if this is still justified today when computers are available.

There are probably many reasons why so many of us still use simple models in spite of the fact that we carry out our calculations on high-capacity computers. It may be called conservatism, but one should realize that there may be many good reasons for being conservative in this case.

(a) Many systems have already been analyzed using simple models. Naturally, one hesitates to do this work all over again.

(b) When analyzing a higher system one would like to base the work on the results already obtained for the lower systems. This is possible only if one retains the model used for a phase in a lower system when treating its extension into a higher system. When such a phase extends all the

way to another lower system, which has not been analyzed previously, it is natural to use the same simple model for the analysis of that system.

(c) Even when a lower system is analyzed separately, it may seem most attractive to use the same simple models that have been used in similar systems, in order to make possible a future combination with other lower systems.

(d) The change to a new model may involve a considerable revision of the calculation programs. If different models are chosen for different phases in a system, one may have to develop different programs for the calculation of different combinations of phases.

(e) An advanced model should be based upon the physical nature of the phase. For different phases it may be desirable to consider different physical effects. In a system with several phases it may thus be necessary to handle a large number of different models and an even larger number of different parameters.

(f) As one goes to more advanced models there may be many alternatives. It is natural to hesitate to develop new calculation programs without knowing if the particular model chosen will be accepted by the scientific community.

(g) The use of different models in different research groups makes cooperation very difficult.

I believe that most of these reasons are still valid today but the situation may soon change. What we need in order to promote the wide-

spread usage of more advanced models is more powerful computer software for the following purposes.

(1) Storing and handling of parameters. We need to store the value of a parameter together with information on the mathematical expressions in which it appears. In order to make the handling flexible it may be necessary to use an advanced data bank system.

(2) Calculation of equilibrium involving different kinds of phases. We need a computer program of considerable generality for the calculation of phase equilibria. It should be capable of handling different models, even such models that have not yet been invented.

(3) Calculation of equilibria under different conditions. For different purposes it may be desirable or even necessary to define the equilibrium in different ways. One often wants to calculate the equilibrium of a system with a given average composition and given temperature and pressure. In other cases one would like to prescribe the value of one or more of the chemical potentials rather than the amounts of the corresponding elements or one would like to prescribe the volume or energy content rather than the pressure or temperature. For a multicomponent phase diagram, one may like to calculate many types of sections.

(4) Optimization of parameter values. The fitting of the thermodynamical models chosen for a system to the experimental information available is usually very tedious and involves much manual work and many subjective decisions. It should be possible to automatize this procedure to a considerable extent.

Although all these areas deserve a detailed discussion, this paper will only deal with the calculation of equilibrium. An attempt will be made to demonstrate that it should be possible to construct computer software by which one can readily calculate the state of equilibrium under many different circumstances.

2. Different methods of calculating phase equilibrium

The state of equilibrium of a system under

constant temperature, pressure and composition is defined by the minimum of the Gibbs energy. It can be found by a numerical procedure. A great number of hill-climbing techniques are available. Some were used long ago for the calculation of the equilibrium in single-phase systems like gases or water solutions. Others have been applied more recently for the calculation of phase equilibria in metallic systems. Using such a method one may construct a program which can accept any models of the individual phases for which one can give an analytical expression for the molar Gibbs energy as a function of composition. It may even be possible to construct the program in such a way that it can accept an analytical expression for the molar Gibbs energy as a function of parameters which cannot be calculated directly from the composition. However, from each set of values for the parameters it must be possible to calculate the composition. One drawback with this method may be the difficulty in finding a numerical procedure which takes advantage of the physical nature of the problem. The calculation may thus be slow and expensive, particularly if many components and phases are involved. Another drawback is that it may be difficult to make this method so general that one may change the equilibrium conditions without modifying the program considerably. As a consequence, one should try to find more powerful methods.

In his classical work Professor Meijering used the criterion that the chemical potential for each component must have the same value in all the phases at equilibrium. This criterion allows a number of equilibrium equations to be formulated and the state of equilibrium is found by solving this system of equations. In most cases this can only be done numerically. It is instructive to see how this criterion is derived from the minimum-of-Gibbs-energy criterion. Let G be the total Gibbs energy of a system containing several phases. G^r is the Gibbs energy of phase r and is a function of the contents of various elements, n_i^r , and also of P and T . In particular, one has

$$dG^r = V^r dP - S^r dT + \sum_i G_i^r dn_i^r, \quad (1)$$

where G'_i stands for $\partial G'/\partial n'_i$. This quantity is called the partial Gibbs energy for element i in the phase r .

Suppose that the total content of each element in the system, n_i , is given. This then restricts the content in the individual phases, n'_i :

$$n_i = \sum_r n'_i. \quad (2)$$

The problem is thus to minimize the function

$$G = \sum_r G^r \quad (3)$$

at constant P and T and under the auxiliary conditions given by eq. (2) when applied to each one of the elements. This problem is solved by the introduction of a Lagrange multiplier, μ_i , for each element. One should form the following function:

$$L = \sum_r G^r + \sum_i \mu_i \left(n_i - \sum_r n'_i \right) \quad (4)$$

and find its minimum with respect to all the variables n'_i . We find

$$\frac{\partial L}{\partial n'_i} = \frac{\partial G^r}{\partial n'_i} - \mu_i = G'_i - \mu_i = 0. \quad (5)$$

Eq. (5) shows that the partial Gibbs energy for element i must have the same value, μ_i , in all the phases. The multiplier μ_i is thus identical to the chemical potential of i in the system. By eliminating the multipliers we obtain the following set of equations:

$$\left. \begin{aligned} G_A^a &= G_A^b = \dots \\ G_B^a &= G_B^b = \dots \\ &\dots \end{aligned} \right\}. \quad (6)$$

Each of the quantities, G_A^a , etc. is a function of the composition parameters and it is possible to solve for the values of all the composition parameters at equilibrium. If one is interested in the values of the chemical potentials of the system, then they can easily be calculated afterwards by inserting the composition values in some of the expressions for the partial Gibbs energies.

The equations are in general nonlinear and the system of equations must be solved numerically.

Usually one starts by guessing how the elements should be distributed over the various phases, and better values are then calculated by iteration using a Newton-Raphson technique. This method of calculation is closely related to the numerical minimization. Some computation time may perhaps be saved because of the use of analytical expressions for the derivatives. On the other hand, more programming may be required.

We started by considering the Gibbs energy of a phase as a function of its contents of various elements, n'_i , and the resulting system of equations contains the partial derivatives with respect to these variables. However, it is more common to work with molar quantities which are obtained by dividing by the number of moles of the phase, n^r :

$$G'_m = G^r/n^r, \quad (7)$$

$$V'_m = V^r/n^r, \quad (8)$$

$$S'_m = S^r/n^r, \quad (9)$$

$$x'_i = n'_i/n^r. \quad (10)$$

The molar Gibbs energy, G'_m , is regarded as a function of the mole fractions, x'_i , and in addition of P and T . One thus has

$$dG'_m = V'_m dP - S'_m dT + \sum_i G'_i dx'_i. \quad (11)$$

One can easily show that the partial Gibbs energy can be calculated from the molar quantity

$$G'_i = G'_m + \partial G'_m / \partial x'_i - \sum_j x'_j \partial G'_m / \partial x'_j. \quad (12)$$

It is thus possible still to use the system of equations given by eq. (6) for calculating the state of equilibrium. However, it is instructive to consider the mathematical analysis of the minimization problem using the molar quantities. We should thus minimize the following function:

$$G = \sum_r n^r G'_m. \quad (13)$$

subject to the previous auxiliary conditions which we can now write as

$$n_i = \sum_r n^r x'_i, \quad (14)$$

but also subject to a new type of auxiliary condition

$$1 = \sum_i x_i^r. \quad (15)$$

In addition to the previous multipliers, μ_i , we now introduce a new type, λ^r , one for each phase. We should form the following function:

$$L = \sum_r n^r G_m^r + \sum_i \mu_i \left(n_i - \sum_r n^r x_i^r \right) + \sum_r \lambda^r \left(\sum_i x_i^r - 1 \right) \quad (16)$$

and find its minimum with respect to all the variables n^r and x_i^r . Now we obtain two types of equations instead of one:

$$\frac{\partial L}{\partial n^r} = G_m^r - \sum_i \mu_i x_i^r = 0, \quad (17)$$

$$\frac{\partial L}{\partial x_i^r} = n^r \frac{\partial G_m^r}{\partial x_i^r} - \mu_i n^r + \lambda^r = 0. \quad (18)$$

It is possible to eliminate the multipliers and the same set of equations will result as before, i.e. eq. (6). However, the fact that there are two types of equations suggests a different strategy. After starting by guessing how the elements should be distributed over the various phases, it may be advantageous to divide the calculation of the state of equilibrium into two steps. In the first step one calculates the values of the chemical potentials of the system using eq. (17), and in the second step one calculates improved values for the compositions of the individual phases using eq. (18). This second step can be carried out for each phase separately, which is a considerable advantage of this method. The whole calculation must be repeated until it has converged sufficiently. When calculating new values of the chemical potentials, μ_i , in one iteration, one would like to use values of the compositions, x_i^r , from the previous iteration. We should thus differentiate eq. (17),

$$dG_m^r = \sum_i \mu_i dx_i^r + \sum_i x_i^r d\mu_i. \quad (19)$$

This can be combined with eqs. (5) and (11) to yield

$$0 = \sum_i x_i^r d\mu_i - V_m^r dP + S_m^r dT \quad (20)$$

which is identical to the Gibbs–Duhem relation. By applying this relation to one iteration step we obtain

$$G_m^r(0) = \sum_i x_i^r(0) \mu_i - V_m^r(0) \Delta P + S_m^r(0) \Delta T, \quad (21)$$

where the left-hand side is identical to $\sum_i x_i(0) \mu_i(0)$. By (0) we indicate that the quantity should be evaluated for the set of values resulting from the previous iteration. By the use of eq. (21) instead of eq. (17) we may calculate P and T in addition to all the μ_i . The number of unknowns is $2 + c$ if c represents the number of components, and the number of equations of the type eq. (21) is p if p represents the number of phases. The calculation can thus be carried out if $c + 2 - p = 0$. This is in agreement with the Gibbs phase rule which defines the degrees of freedom, f , as

$$f = c + 2 - p. \quad (22)$$

From the form of eq. (21) we can see that if $f > 0$, one may specify the values of f of the variables μ_i , P and T and calculate the remaining ones.

Since eq. (20) is identical to the Gibbs–Duhem relation, it is evident that the calculation, which has now been described, is not restricted to cases where the state of equilibrium is defined by the minimum of the Gibbs energy of the system. In fact, after minor modifications this method of calculation can be used under a large number of different conditions, as will be described below. It should further be noticed that the Gibbs–Duhem relation can be further generalized by including other intensive variables, for instance the electrical potential.

The method of dividing the calculation into two steps and thus limiting the number of unknown variables in each calculation was successfully tried by Eriksson [1, 2]. The method described in the present paper may be regarded as a generalization of his method. The basic principles have already been described in two papers [3, 4] and used in the construction of a program for the calculation of phase diagrams [5]. The

present description goes further on some points.

It should be emphasized that it is in principle possible to apply a Newton–Raphson technique for solving the whole system of equations formed by eqs. (14), (15), (17) and (18) without linearizing and without dividing into different steps. This may be a convenient way of handling the calculation when there are not too many variables, but it may not be very practical in large systems with many components and several phases.

3. Modification to different equilibrium conditions

μ_i , P and T are all intensive properties. The corresponding extensive properties are n_i , V and S . When p and c are specified in such a way that $f = 0$, then one may in addition specify the values of these extensive properties. This does not affect the calculation of the state of equilibrium. Afterwards one may use the specified values in order to evaluate the amount of each phase, n' , by solving the following system of equations:

$$n_i = \sum_r n' x'_i, \quad (23)$$

$$V = \sum_r n' V'_m, \quad (24)$$

$$S = \sum_r n' S'_m. \quad (25)$$

The values of the left-hand sides are specified and the molar quantities on the right-hand sides can be evaluated once the state of equilibrium has been found. If all the calculated n' are not positive, then the system has been chosen outside the chosen phase field and one must change either one of the choices.

If $f > 0$ and one has specified the values of f intensive variables, then one should not specify the values of the f corresponding extensive properties. For the calculation of the n' parameters there will thus be $c + 2 - f$ equations which is equal to p , the number of n' parameters. Alternatively, one may decide not to specify the values of the intensive variables but instead to let the specification of the corresponding extensive variables influence the calculation of the state of equilibrium and thus

influence the values of the intensive variables. That type of calculation will be more complicated. It can make use of the differentiated form of eqs. (23)–(25):

$$dn_i = \sum_r x'_i dn' + \sum_r n' dx'_i, \quad (26)$$

$$\begin{aligned} dK &= \sum_r K'_m dn' + \sum_r n' dK'_m \\ &= \sum_r K'_m dn' + \sum_r n' \frac{\partial K'_m}{\partial P} dP + \sum_r n' \frac{\partial K'_m}{\partial T} dT \\ &\quad + \sum_r \sum_i \frac{\partial K'_m}{\partial x'_i} dx'_i. \end{aligned} \quad (27)$$

Here K stands for V and S . When applying these equations to the iterative procedure we obtain:

$$n_i + \Delta n_i = \sum_r x'_i(0) n' + \sum_r n'(0) \Delta x'_i, \quad (28)$$

$$\begin{aligned} K + \Delta K &= \sum_r K'_m(0) n' + \sum_r n'(0) \frac{\partial K'_m(0)}{\partial P} \Delta P \\ &\quad + \sum_r n'(0) \frac{\partial K'_m(0)}{\partial T} \Delta T \\ &\quad + \sum_r \sum_i \frac{\partial K'_m}{\partial x'_i} \Delta x'_i, \end{aligned} \quad (29)$$

where n_i stands for $x'_i(0)n'(0)$, i.e. the chosen i content of the system. Δn_i is usually zero but can be used to represent a sudden change made to the system. K and ΔK have corresponding definitions. In addition to the intensive variables which are already present in eq. (21), these new equations contain n' and $\Delta x'_i$. In order to make these equations useful, one must eliminate the $\Delta x'_i$ variables, and that can be done by means of the equations used in the second step of the calculation of the state of equilibrium.

In the second step of the calculation one can formulate separately a system of equations for each phase by applying eqs. (15) and (18). The equations obtained from eq. (18) are generally nonlinear. The system of equations can be solved numerically by a Newton–Raphson technique each time one has calculated a new set of values for the intensive variables. However, in order to eliminate $\Delta x'_i$ from eqs. (28) and (29) we must find analytical expressions for $\Delta x'_i$ as functions of the intensive variables. This can be done by

linearizing eq. (18) using the fact that G_m is a function of P , T and x_i^r :

$$d\left(\frac{\partial G_m^r}{\partial x_i^r}\right) = \frac{\partial V_m^r}{\partial x_i^r} dP - \frac{\partial S_m^r}{\partial x_i^r} dT + \sum_j \frac{\partial^2 G_m^r}{\partial x_i^r \partial x_j^r} dx_j^r. \quad (30)$$

By applying this expression to eq. (18) in the iterative procedure we obtain

$$\begin{aligned} \mu_i - \frac{\partial G_m^r(0)}{\partial x_i^r} - \frac{\partial V_m^r(0)}{\partial x_i^r} \Delta P + \frac{\partial S_m^r(0)}{\partial x_i^r} \Delta T \\ = \sum_j \frac{\partial^2 G_m^r(0)}{\partial x_i^r \partial x_j^r} \Delta x_j^r + \frac{\lambda^r}{n^r}. \end{aligned} \quad (31)$$

Eq. (15) yields

$$0 = \sum_j \Delta x_j^r. \quad (32)$$

This system of linear equations can be solved analytically and will give all the Δx_j^r as linear functions of the intensive variables. Let us represent these expressions with

$$\Delta x_i^r = e_{ij}^r + \sum_j e_{ij}^r \mu_i + e_{ij}^r \Delta P + e_{ij}^r \Delta T. \quad (33)$$

The expressions for Δx_j^r can be inserted in eqs. (28) and (29) which will thus become linear functions of the intensive variables and n^r . Together with the equations of the type given by eq. (21) we thus have a system of $p + c + 2$ equations and the same number of unknowns when no intensive variable has been specified. It can be solved analytically. After the intensive variables have been calculated, eq. (33) will be used to calculate numerical values of all the Δx_i^r .

If the value of g of the intensive variables are specified, and $g < f$, then one must specify the values of exactly $c + 2 - g$ extensive properties. We thus have a system of $p + c + 2 - g$ equations and the same number of unknowns (p of type n^r and $c + 2 - g$ intensive variables). As already emphasized, the calculation is greatly simplified if $g = f$ because the remaining intensive variables can then be calculated directly from the system of equations given by eq. (29) without combination with the equations which contain extensive variables.

So far we have considered the possibility of

specifying the values of the extensive properties which appear in eqs. (23)–(25). Alternatively, one may specify the value of some other quantity. Instead of V or S one may choose some other extensive thermodynamic property, e.g. the internal energy U . For any such property, K , we obtain

$$K = \sum_r n^r K^r. \quad (34)$$

Instead of specifying the values of all of n_i , one may specify one or more of the n^r . In such a case one adds a new equation and changes one of the old ones in the system of eqs. (23)–(25). If the value of n^a is specified instead of n_A then one obtains:

$$a = n^a, \quad (35)$$

$$0 = \sum_r n^r x_A^r - n_A. \quad (36)$$

One may calculate a phase boundary by putting a to zero. Eq. (36) yields the following modified version of eq. (28):

$$0 = \sum_r x_A^r(0) n^r + \sum_r n^r(0) \Delta x_A^r - n_A. \quad (37)$$

Sometimes one would like to calculate a section through a phase diagram along a straight line in composition. The straight line can be defined by $c - 1$ linear equations of the type

$$0 = \sum_i b_i^k n_i. \quad (38)$$

When these equations are introduced, one should also modify eq. (28) for all the components

$$0 = \sum_r x_i^r(0) n^r + \sum_r n^r(0) \Delta x_i^r - n_i. \quad (39)$$

In order to calculate where a phase boundary intersects this section, one can also introduce, as a new equation,

$$0 = n^a, \quad (40)$$

where a is the phase that disappears on the phase boundary. We have thus added c linear equations and c unknowns (all of the n_i).

It may also be of interest to specify the composition of some phase. For each x_i^r which is specified, the corresponding Δx_i^r is put to zero. If, for instance, Δx_A^{α} is put to zero then eq. (33) yields

$$-e_{0A}^{\alpha} = \sum_i e_{iA}^{\alpha} \mu_i + e_{PA}^{\alpha} \Delta P + e_{TA}^{\alpha} \Delta T. \quad (41)$$

This equation must now be added to the system of equations given by eq. (21). It is thus necessary to decrease the number of specified intensive or extensive variables by one.

4. Changes in the set of phases taking part in the equilibrium

So far we have assumed that the set of phases taking part in the equilibrium is given from the beginning. However, it was mentioned that some of the n^r parameters may be found to be negative if the overall composition of the system has also been chosen from the beginning. Some change must then be made in the choice. If the overall composition is not to be changed, then one should exclude a phase from taking part in the equilibrium when its n^r goes negative. This can be done by omitting the corresponding Gibbs-Duhem relation, eq. (21), and also excluding this phase from the summations over r in eqs. (28) and (29).

It should also be recognized that some phase, α , may be missing in the original guess of which phases should take part in the equilibrium. This can be tested by evaluating the supersaturation or undersaturation with respect to α . A difficulty is that such a test must be carried out for the most favourable compositions of α . One method of solving this problem is continuously to calculate the most favourable composition of α . This can be done by applying a hypothetical pressure P^{α} to the α phase, which would bring this phase into equilibrium with the other phases. That pressure can be evaluated from eq. (21) each time the intensive properties have been evaluated,

$$V_m^{\alpha}(0) \Delta P^{\alpha} = \sum_i x_i^{\alpha}(0) \mu_i + S_m^{\alpha}(0) \Delta T - G_m^{\alpha}(0), \quad (42)$$

where ΔP^{α} stands for $P^{\alpha} - P^{\alpha}(0)$. After evaluat-

ing ΔP^{α} one can calculate the most favourable composition of α from eqs. (31) and (32) by inserting ΔP^{α} as ΔP . Provided the system is undersaturated with respect to α the value of P^{α} will be lower than the value of P for the phases in equilibrium. When $P^{\alpha} - P$ turns positive, the phase should be included in the equilibrium unless some nucleation barrier is assumed. In that case one can evaluate the size of a critical nucleus from

$$r = 2\sigma/(P^{\alpha} - P) \quad (43)$$

and maybe use some special criterion in order to decide when α should start to form. When α is first included in the equilibrium one adds the corresponding Gibbs-Duhem relation and extends the summations over r in eqs. (28) and (29) to include the α phase. In the first cycle of iteration $n^{\alpha} = 0$.

There are also cases where one would like to consider a phase equilibrium where one phase is under a higher pressure than another, owing to the action of the surface tension in a curved phase interface.

It should be noted that (0) here stands for the values obtained for the α phase in the previous iteration. It may sometimes be convenient to calculate V_m^{α} , S_m^{α} , G_m^{α} and their derivatives at the value of P which holds in the phases which are in equilibrium. As an approximation one may often introduce these values in the equations, but should then interpret ΔP^{α} as $P^{\alpha} - P(0)$.

When calculating a two-dimensional phase diagram or section through a phase diagram one wants to find the shape of various phase boundaries and, in particular, their intersections. In order to minimize the cost of calculation it may be advantageous to try to go directly to the next intersection. Suppose that one has just calculated a point on a boundary. The direction of the boundary is then given by the set of Gibbs-Duhem relations, eq. (21), one for each participating phase, and with the thermodynamic properties evaluated at the calculated point. It can be calculated directly from this set of equations if the diagram is plotted with intensive variables on the axes. In order to find where this boundary intersects those phase boundaries in-

volving another phase, α , one should include the Gibbs–Duhem relation for that phase and in the first iteration cycle one should add a term $V_m^\alpha(0)(P(0) - P^\alpha(0))$ to $G_m^\alpha(0)$ if the notation (0) is used to identify the conditions of that phase under the hypothetical pressure. The extra term should not be added if one uses the approximate treatment where the thermodynamic properties of the α phase are evaluated at the pressure of the phases in the equilibrium. The point of intersection is then obtained directly if the compositions of all the phases are constant. If they vary one must go through some iteration cycles, but the first cycle will generally give approximately correct results.

After having calculated the point of intersection, one must check that there is no other phase which has a pressure higher than that of the phases in the equilibrium. If this is the case, then one should exchange the newly added α phase with that phase. In order to have the best chance of trying the right phase first, one should choose the phase which has the highest pressure.

The calculation will be more complicated if the diagram is a section which is defined using extensive variables. One would then have to use the whole system of equations which was used to calculate the first point on the phase boundary. When adding eq. (21) for the new phase α one should also add the equation

$$0 = n^\alpha. \quad (44)$$

A similar procedure can be used to find a phase boundary if one starts inside a one-phase field.

A phase boundary sometimes ends in a critical point where two phases become one. An indication that this may be the case is given by the calculation of the point of intersection with another phase which does not converge but may show, during the iteration, that the two phases approach each other. The critical point can then be located by a step-wise movement in the phase diagram or by a direct calculation using the special criteria of a critical point.

5. The application of different thermodynamic models

The fact that the calculation of the state of

equilibrium can be divided into two steps and that the calculation for each phase is carried out separately from all the others, opens the possibility of using different models for the phases without complicating the calculation. For a new model which gives the Gibbs energy as a function of composition it is necessary only to give the analytical expressions for the various thermodynamic quantities.

Often one may obtain simpler expressions by using a different set of parameter values than the ordinary mole fractions. For a phase which has more than one sublattice it may be more convenient to use site fractions, i.e. mole fractions defined for each sublattice. Let us denote them by $y_i^{\alpha s}$, and let $a^{\alpha s}$ be the number of sites on the sublattice s in one mole of phase α . We can always calculate the mole fractions from the set of site fractions. With our definitions we find

$$1 = \sum_i y_i^{\alpha s}, \quad (45)$$

$$1 = \sum_s a^{\alpha s}, \quad (46)$$

$$x_i^\alpha = \sum_s a^{\alpha s} y_i^{\alpha s}. \quad (47)$$

In order to make the mathematics simple it is advantageous to define $a^{\alpha s}$ with respect to one mole of lattice sites in the phase, eq. (46). If some sites are vacant, one must then include the vacancies in the summation over i in eq. (45). The total number of moles of real atoms will thus be less than unity in one mole of the phase.

Two important cases should now be discussed. The first case can be illustrated by the phase $(A, B)_{a1} (C, D)_{a2}$, where A and B occupy one sublattice and C and D another. There are four components but the composition can be varied by two degrees of freedom only. One may, for instance, regard y_B^1 and y_D^2 as the independent variables, whereas $y_A^1 = 1 - y_B^1$ and $y_C^2 = 1 - y_D^2$. In this case it is impossible to define the partial Gibbs energy for an element because one cannot keep the other contents constant when one is varied. The strategy which leads to eq. (6) must therefore be modified. On the other hand, the new strategy can be used with a very slight change. First, the auxiliary condition for this

phase, eq. (15), should be replaced by one for each sublattice, as given by eq. (45). It is thus necessary to introduce one multiplier λ^{as} for each sublattice. This change will not affect the derivation of eq. (17) and the first step of the calculation can thus be carried out as before. On the other hand, when deriving the equation corresponding to eq. (18) one must replace x_i^a in the μ -term of eq. (16) with the expression given by eq. (47), and one should for the last term write $\sum_s \lambda^{as} (\sum_i y_i^{as} - 1)$. As a consequence, one must take the derivative with respect to y_i^{as} instead of x_i^a :

$$\frac{\partial L}{\partial y_i^{as}} = n^a \frac{\partial G_m^a}{\partial y_i^{as}} - \mu_i n^a a^{as} + \lambda^{as} = 0. \quad (48)$$

In the second step of the calculation one should now solve the system of equations formed by eqs. (47) and (48). By linearizing we obtain

$$a^{as} \mu_i - \frac{\partial G_m^a}{\partial y_i^{as}} = \sum_j \frac{\partial^2 G_m^a}{\partial y_i^{as} \partial y_j^{as}} \Delta y_j^{as} + \frac{\lambda^{as}}{n^a}, \quad (49)$$

$$0 = \sum_j \Delta y_j^{as}. \quad (50)$$

When solving this system of equations we find the following type of relations, instead of eq. (33):

$$\Delta y_j^{as} = e_{ij}^{as} + \sum_s a^{as} \left[\sum_i e_{ij}^{as} \mu_i + e_{pj}^{as} \Delta P + e_{Tj}^{as} \Delta T \right] \quad (51)$$

These expressions can be inserted into eq. (28) because Δx_j^a (in eq. (28) denoted by Δx_i^a) can be calculated from Δy_j^{as} using eq. (47):

$$\Delta x_j^a = \sum_s a^{as} \Delta y_j^{as}. \quad (52)$$

In the case which has just been considered it is possible not only to express each one of the x_i^a as a function of the set of y_i^{as} but also to express each one of the y_i^{as} as a function of the set of x_i^a . The state of the phase is thus defined if the composition is given. The other important case that should be discussed is found when there may be vacancies on two sublattices or vacancies on one sublattice at the same time as one element on the same sublattice can go into another one or, finally, two elements can go into the same two sublattices. In this case the state of the

phase is not defined if the composition is given. In principle, for each given composition of the phase it is necessary to find the distribution of the elements by minimizing the Gibbs energy. It is interesting that this complication can be handled without any further modification of the method of calculation, as shown recently in a case of two sublattices with vacancies on both [6]. It is of course possible to investigate the properties of such a phase by a separate investigation and thus to explore how the state actually changes with composition. However, one cannot hope to describe such results in closed form. A separate investigation may thus be of little value.

A similar problem is found in phases where atoms can form molecular aggregates. By introducing the notation k for the species, we find

$$1 = \sum_k y_k^{as}, \quad (53)$$

where the summation is carried out over molecules as well as separate atoms, depending upon how the Gibbs energy of the phase is described. In many cases an element may occur in two states of ionization and they must be treated as different species. The number of i atoms in one mole of the phase is given by

$$x_i^a = \sum_s a^{as} \sum_k b_i^k y_k^{as}, \quad (54)$$

where b_i^k gives the number of i atoms per unit of the k species. It should be noted that the sum of x_i^a over all the real atoms is rarely equal to unity if a^{as} is defined for one mole of species, counting vacancies as a separate species.

In ionic phases one may also want to include a requirement of electroneutrality which may be expressed by means of the valency v_k :

$$0 = \sum_s a^{as} \sum_k v_k y_k^{as}. \quad (55)$$

For this new auxiliary condition one should introduce a new multiplier λ^{ac} . When forming the L function, corresponding to eq. (16), we should now replace x_i^a in the μ -term by the expression given by eq. (54). We should replace $\lambda^a (\sum_i x_i^a - 1)$ in the λ -term by $\sum_s \lambda^{as} (\sum_k y_k^{as} - 1)$ and we should add a term $\lambda^{ac} \sum_s a^{as} \sum_k v_k y_k^{as}$. When

deriving an equation corresponding to eqs. (18) or (48) we will now obtain:

$$\frac{\partial L}{\partial y_k^{\alpha s}} = n^{\alpha} \frac{\partial G_m^{\alpha}}{\partial y_k^{\alpha s}} - \sum_i \mu_i n^{\alpha} a^{\alpha s} b_i^k + \lambda^{\alpha s} + \lambda^{\alpha e} a^{\alpha s} v_k = 0. \quad (56)$$

A related problem is the case of short range order caused by a tendency to form more unlike nearest neighbour bonds than should be found if the atoms were mixed at random. Many advanced models have been developed for this phenomenon, the simplest being the quasichemical theory. The model works with the fractions of various types of bonds, x_{ij}^{α} ,

$$1 = \sum_i \sum_j x_{ij}^{\alpha}. \quad (57)$$

The composition of the phase can be directly calculated from these fractions. For an element t we find

$$x_t^{\alpha} = \sum_j \frac{1}{2} x_{jt}^{\alpha} + \sum_i \frac{1}{2} x_{it}^{\alpha}.$$

When forming the L function one should introduce this expression into the μ -term; the λ -term then becomes $\lambda^{\alpha} (\sum_i \sum_j x_{ij}^{\alpha} - 1)$. One thus obtains:

$$\frac{\partial L}{\partial x_{ij}^{\alpha}} = n^{\alpha} \frac{\partial G_m^{\alpha}}{\partial x_{ij}^{\alpha}} - \frac{1}{2} \mu_i n^{\alpha} - \frac{1}{2} \mu_j n^{\alpha} + \lambda^{\alpha} = 0. \quad (58)$$

It seems possible that many more advanced models could be handled just as easily. Other

cases of ordering are found in magnetic materials.

6. Concluding remarks

It has been demonstrated that one should be able to construct a very general program for the calculation of phase equilibrium. There is also a great need for general programs for the handling of data and models and, in particular, for the optimization of parameter values to the best agreement with experimental information. The progress in the field of coupling between thermochemistry and phase diagrams depends critically upon the development of such general programs.

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